

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/10/2026	DATE SUBMITTED:	09/10/2026
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ACTIVITY NO. 7.1	Hands-on Activity 7.1: Managing Files and Creating Roles in Ansible
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1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it "files." Create a file inside that directory and name it "default_site.html." Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
</head>
<body>
<h1>Welcome to My Web Server</h1>
<p>This webpage was deployed using Ansible.</p>
<p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

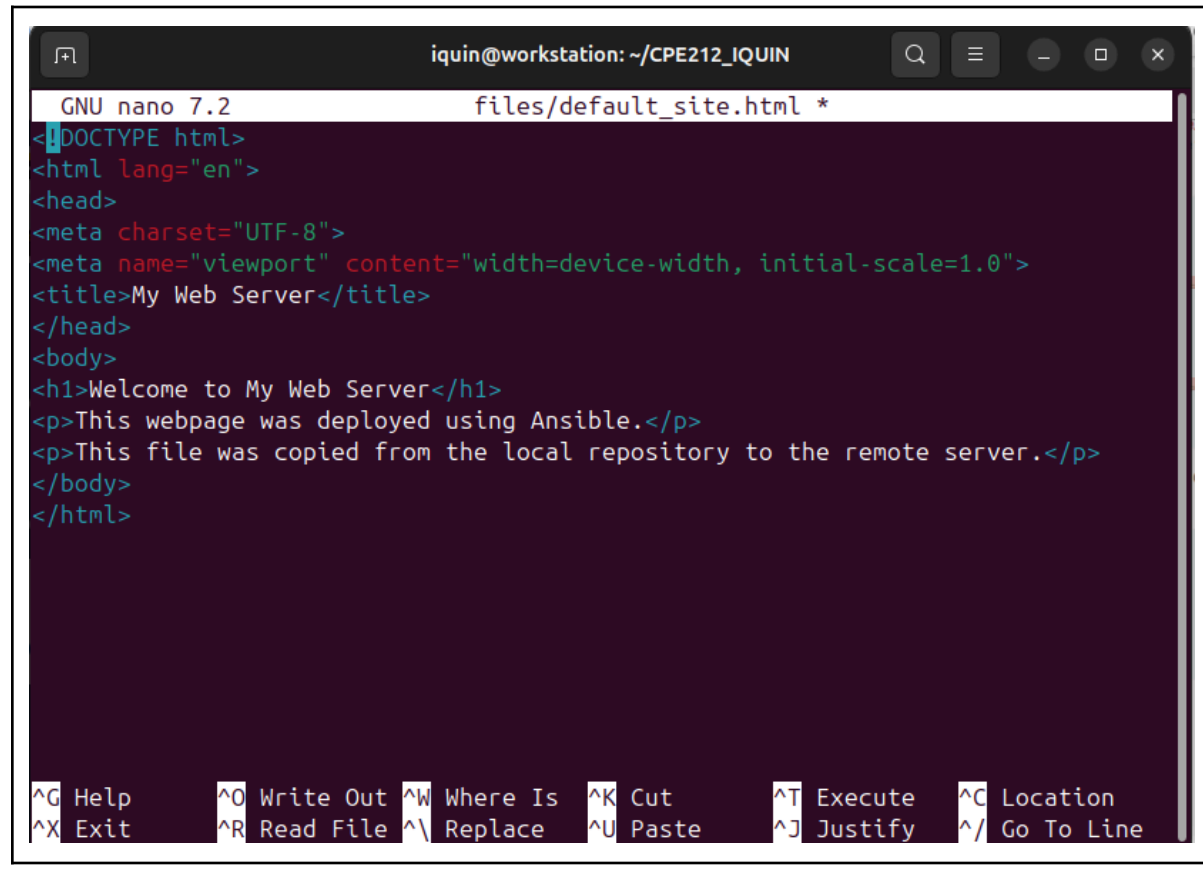
```
- name: copy default html file for site
tags: apache, apache2, httpd
copy:
src: files/default_site.html
dest: /var/www/html/index.html
owner: root
group: root
mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

None

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

5. Sync your local repository with GitHub and describe the changes.



```

iakin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 files/default_site.html *
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
</head>
<body>
<h1>Welcome to My Web Server</h1>
<p>This webpage was deployed using Ansible.</p>
<p>This file was copied from the local repository to the remote server.</p>
</body>
</html>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
  
```

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 site.yml
- name: copy default html file for site
  tags: apache, apache2, httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

[Wrote 8 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/ Go To Line

```

iquin@workstation: ~/CPE212_IQUIN
TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

TASK [copy default html file for site] *****
changed: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [database : install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [database : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY RECAP *****
192.168.56.121      : ok=3    changed=2    unreachable=0    failed=0    s
skipped=2    rescued=0    ignored=0
192.168.56.122      : ok=3    changed=1    unreachable=0    failed=0    s
skipped=0    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml -K
BECOME password:

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for Ubuntu servers] *****
changed: [192.168.56.121]

TASK [webserver : install apache and php for CentOS servers] *****
skipping: [192.168.56.121]

TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

TASK [copy default html file for site] *****
changed: [192.168.56.121]

```

```

iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 site.yml
---
- hosts: workstations
  become: true
  tasks:
    - name: install unzip
      package:
        name: unzip
    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
        dest: /usr/local/bin
        remote_src: yes
        mode: '0775'
        owner: root
        group: root

- hosts: web_servers
  become: true
  roles:
    - role: webserver

[ Read 36 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

```

```
iquin@workstation: ~/CPE212_IQUIN
TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

TASK [copy default html file for site] *****
ok: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [database : install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [database : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY RECAP *****
192.168.56.121 : ok=3    changed=0    unreachable=0    failed=0    s
skipped=2    rescued=0    ignored=0
192.168.56.122 : ok=3    changed=1    unreachable=0    failed=0    s
skipped=0    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$
```

```
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml -K
BECOME password:

PLAY [all] *****
*****

TASK [Gathering Facts] *****
*****
ok: [192.168.56.121]
ok: [192.168.56.122]
ok: [192.168.56.123]

TASK [update repository index (CentOS)] *****
*****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
*****
skipping: [192.168.56.122]
ok: [192.168.56.121]
```

```

iquin@workstation: ~/CPE212_IQUIN
*****
skipping: [192.168.56.121]

TASK [web_servers : copy default html file for site] **
*****
ok: [192.168.56.121]

PLAY [db_servers] *****
*****

TASK [Gathering Facts] *****
*****
ok: [192.168.56.122]

TASK [db_servers : install mariadb package (CentOS)] **
*****
ok: [192.168.56.122]

TASK [db_servers : Mariadb- Restarting/Enabling] *****
*****
changed: [192.168.56.122]

PLAY [file_servers] *****
*****

PLAY RECAP *****
*****
192.168.56.121      : ok=7      changed=1    unrea
chable=0    failed=0    skipped=4    rescued=0    ignor
ed=0
192.168.56.122      : ok=7      changed=1    unrea
chable=0    failed=0    skipped=2    rescued=0    ignor

```

```
site.yml --K
iquin@workstation:~/CPE212_IQUIN$ cat -n site.yml
 1  ---
 2  - hosts: all
 3    become: true
 4    pre_tasks:
 5      - name: update repository index (CentOS)
 6        tags: always
 7        dnf:
 8          update_cache: yes
 9          changed_when: false
10        when: ansible_distribution == "CentOS"
11
12      - name: install updates (Ubuntu)
13        tags: always
14        apt:
15          update_cache: yes
16          changed_when: false
17          when: ansible_distribution == "Ubuntu"
18
19  - hosts: all
20    become: true
21    roles:
22      - base
23
24  - hosts: workstations
25    become: true
26    roles:
27      - workstations
```

```

iquin@workstation: ~/CPE212_IQUIN
24 - hosts: workstations
25     become: true
26     roles:
27         - workstations
28
29 - hosts: web_servers
30     become: true
31     roles:
32         - web_servers
33
34 - hosts: db_servers
35     become: true
36     roles:
37         - db_servers
38
39 - hosts: file_servers
40     become: true
41     roles:
42         - file_servers
iquin@workstation:~/CPE212_IQUIN$ cat -n hosts
 1 [web_servers]
 2 192.168.56.121 ansible_user=iquin ansible_ssh_p
ass=Qwerty@123
 3
 4 [db_servers]
 5 192.168.56.122 ansible_user=melquisedec ansible
_ssh_pass=Qwerty@123
 6
 7 [file_servers]
 8 192.168.56.123 ansible_user=iquin ansible_ssh_p
ass=Qwerty@123
iquin@workstation:~/CPE212_IQUIN$

```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following:

1. What is the importance of creating roles?
 - The creation of roles is important because it organizes the related tasks, files, and the variable into reusable like the activity. Basically making it short like the site yaml which makes it look clean and easier to read and actually understand.
2. What is the importance of managing files?
 - So managing files properly is really important because this ensures that the command you are using can find the respective place where that file is. Its also to ensure that every server receives consistent content.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

- *Quill bot, gemini*

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

- *The main purpose of the use is mainly for grammatical errors and checking if my indentations are correct or misplaced*

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.